MECHANICAL ENGINEERING — STANFORD UNIVERSITY, CLASS OF 2028

Alex Hong

I study mechanical engineering at Stanford. In summer 2026 I worked on carbon nanotube synthesis in the Boies Lab. I like projects I can put my hands on: a purely mechanical launcher, a stage built for a party, and an aluminum piano I'm making for ME 103.

ALEXJYH@STANFORD.EDU
LINKEDIN.COM/IN/ALEXJHONG
ALEXHONGSTA.GITHUB.IO — 3D MODELS AND CAD FILES

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Custom Aluminum Piano, exploded: frame, glass window, 20 keys on the pivot rod, base.

01

Boies Lab

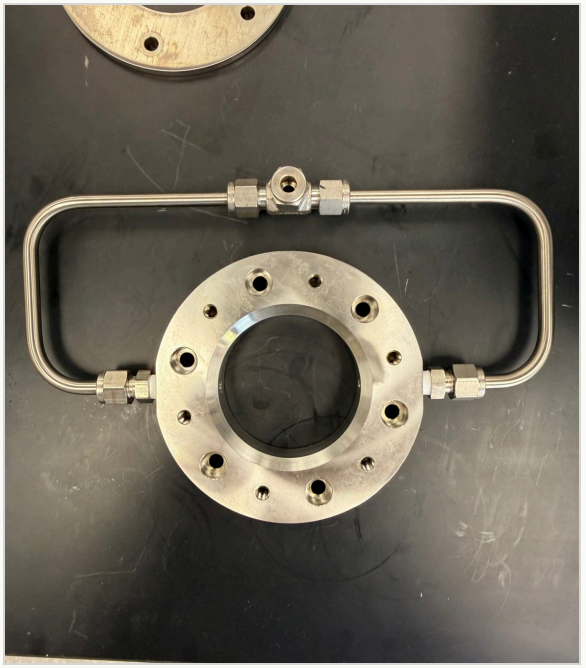
SUMMER UNDERGRADUATE RESEARCHER
JUNE - SEPT 2026
STANFORD SCHOOL OF ENGINEERING

- I tested injector configurations in a floating-catalyst CVD reactor that runs at 1300 °C. Counterflow injection, versus main-flow, raised carbon nanotube production to 216 mg/hr.
- Found the optimal injection point, 28.3 cm, through repeated experimental runs.
- Designed the custom quartz-tube flange assemblies in Fusion 360: sealing geometry and injector-tube integration for high-temperature service.
- Worked alongside a PhD collaborator, iterating reactor geometry against experimental results.

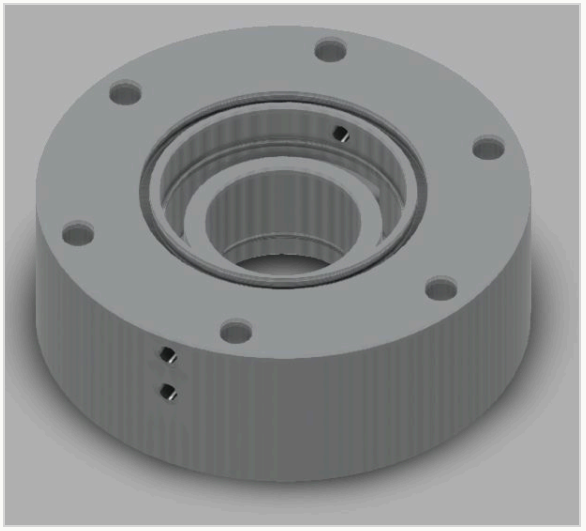
3D MODEL + STEP FILE:
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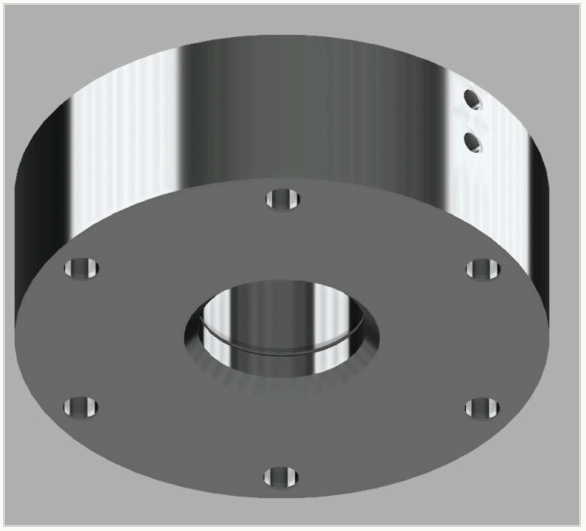
Flange assembly on the reactor, injector line at top.



Flange with injector feed loop, on the bench.



Counterflow flange, Fusion 360 model.



Counterflow flange, underside.

02

Custom Aluminum Piano

ME 103 · FALL 2026 · IN PROGRESS

A 20-key MIDI keyboard, E to B (12 white keys, 8 black), made entirely of 356A aluminum. The frame, base, and key blanks are sand cast in the PRL foundry from 3D-printed PLA patterns, then finished on a manual mill. The keys pivot on a Ø.250 in stainless rod, and a glass window sits in the back wall.

All 12 white keys are cast and machined as one 11 in bar, and the 8 black keys as one 4.2 in bar, so every key gets the same profile. The bars are sawed apart afterward. All 20 pivot holes are drilled in one setup so they line up on the rod.

Writing the operations sequence turned up five things to fix in CAD before printing patterns. The enclosed glass slot can't be sand cast, so it became an open rabbet. The top skin was .044–.056 in thick, too thin for the casting to fill. Both pivot holes were blind, so the rod couldn't go in. The key stack had no end clearance, and the screws reached only about .10 in into the frame.

The BOM comes to $104.84 against a $130 budget. Electronics are a later phase.

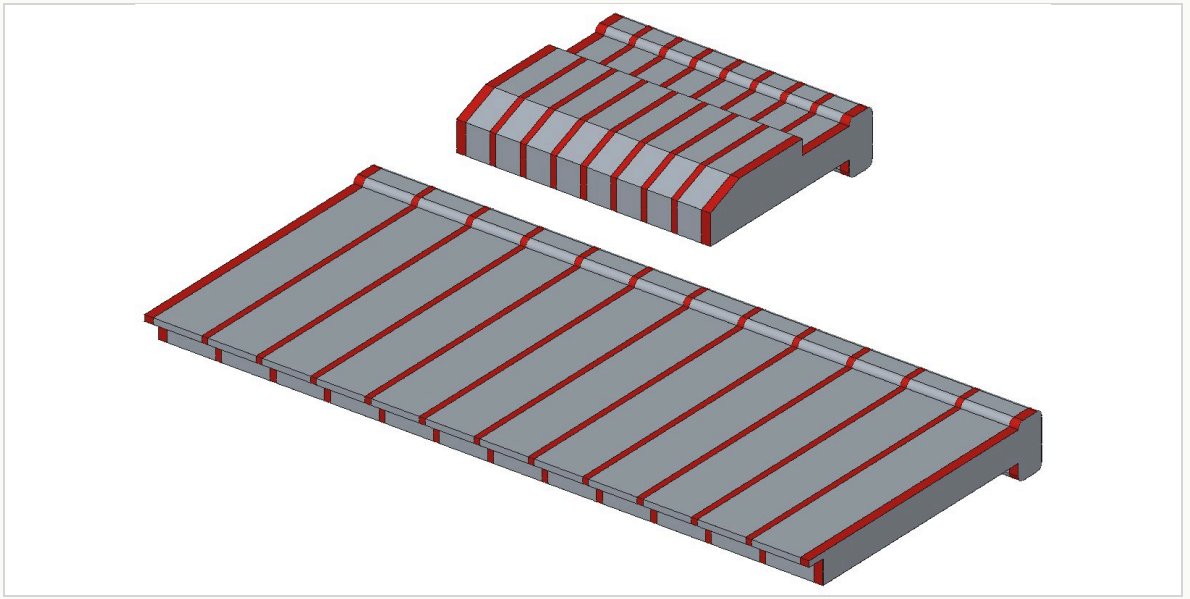
OPERATIONS SEQUENCE, BOM, 3D MODEL, STEP FILE:
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Assembly, front.



Assembly, back, with the glass window.



OP 40: sawing the cast bars into keys at .900 and .500 in pitch. Red is material removed.

03

Spring-Powered Ping Pong Launcher

ME 102 PRODUCT REALIZATION · WINTER 2026

A purely mechanical, spring-driven launcher inside a 12 × 12 × 24 in envelope. It fired through a target board at 5–7 ft over a two-minute run.

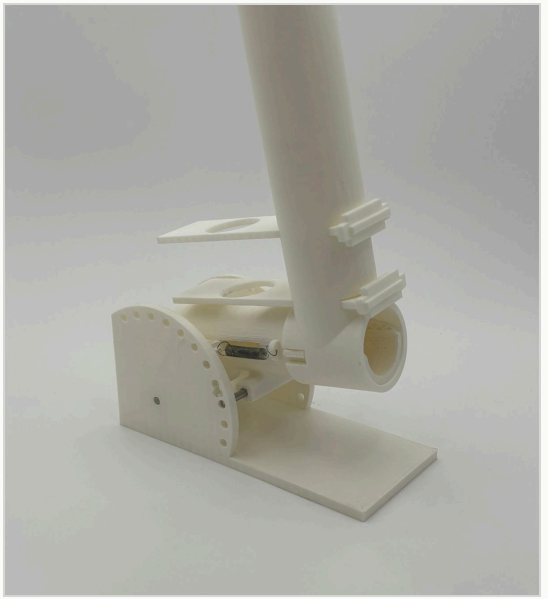
The scored design metric was the static counterweight resisting launch reaction forces. I cut it with geometry instead of added mass.

The paper trail: free-body diagrams, force and moment balances, stored-energy calculations, and a dimensioned drawing package with tolerances, BOM, and exploded views. Parts were laser-cut and 3D-printed. The Fusion assembly has six parts: base, barrel, funnel, slider, slot, and shaft.

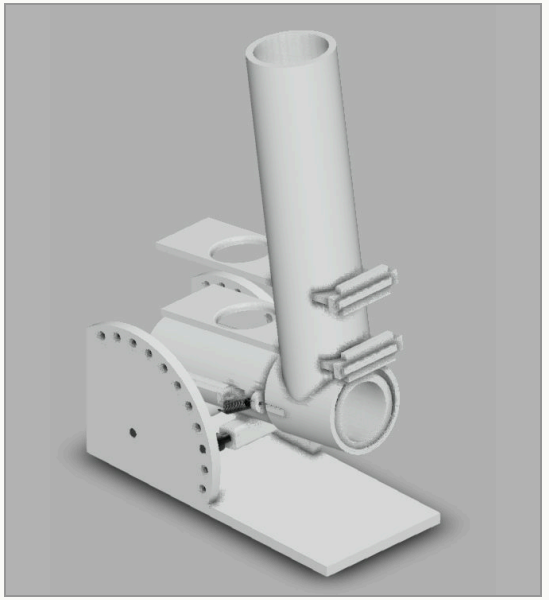
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Built launcher.



Side plate and spring.



Fusion 360 assembly.

04

Outdoor Stage Build

SIGMA NU SUNSPLASH · 2026 · ~1,340 RSVPS

A wooden stage, DJ table, and tiered platforms for Sigma Nu's Sunsplash. I modeled it in Fusion 360: 17 platforms in four sizes (a 2 ft stage, 1 ft and standard die tables, and 4 ft die tables with rails), 8 fence sections, and the DJ table.

I sized the 2×4, 4×4, and plywood members from that model. Then we bought the lumber and built it on site.

The aluminum truss over the DJ table was bought, not built. I DJ'd the party from the table.

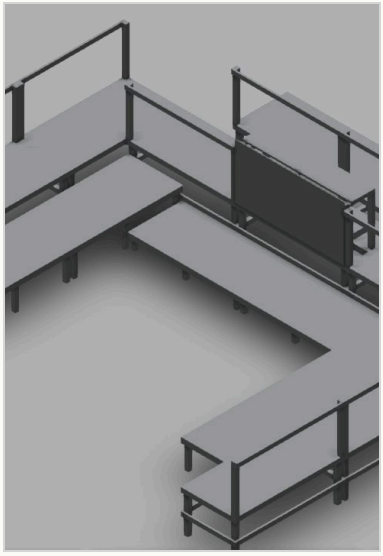
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Stage at night, mid-paint.



DJing the party from the table.



Fusion 360 layout.



Framing on site, before paint.



One die table, plywood on a braced frame.